

Course Management System

Piyush Sawarkar
22M0772

Kiran kumar Kunche
22M0755

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1 Description

Defines an education management information system (EMIS) as a data system that collects, monitors, manages, analyzes, and disseminates information about education inputs, processes, and outcomes—in particular, student learning. A successful EMIS depends on the interplay of appropriate policies, budget, human resources, organizational structure, and institutions to produce valid education data.

It should empower all data users to benefit from the data and foster data-driven decision making, transparency, and accountability. Data users can include central and local authorities, schools, teachers, parents, and the broader community, including media, researchers, and nongovernmental organizations (NGOs).

A smart system requires (1) a high degree of reliability, effi-

ciency, and sustainability—of both the structure and the system; (2) full integration of all functions of the system; (3) high security of the infrastructure, particularly when subjected to extreme and unconventional conditions; (4) continuous health and integrity monitoring; (5) damage detection and self-recovery; and (6) an intelligent operational and management system.

2 Achievements

A *typical* user for this EMIS are two. One the *Course Instructor* and the *Student*. Any user has to initially go through a sophisticated procedure to authenticate himself/herself before accessing this EMIS.

3 Course Instructor as a user

A Course Instructor is referred as an "Admin" in this system. Typical

Rights for such a user for this system has to view the *TOP 10* performers in any of the courses offered in this Fall. A total of 5 courses are offered this Fall, namely, Design and Engineering of Computing Systems abbreviated as *DECS*, Foundations of Machine Learning abbreviated as *FML*, Information Retrieval and Database Management System abbreviated as *DBMS*, Algorithms and Software Lab. For each of the above courses the system lists the top performers.

4 Student as a user

A student is offered numerous options once authenticated, namely, to checkout the courses in which he/she has enrolled for in that Fall, ability to checkout Calendar, access a statically built to-do list and to checkout his/her Time Table for that Fall.

For every course student has the ability to see a detailed analytical report regarding not just his/her performance in that course but also a *comparative* performance analysis report amongst all course participants. Various Plotting tools available in Javascript are employed for achieving the above.

5 Future Work

This EMIS holds a tremendous potential for an extensive future work.

For the *Course Instructor* as a User: The ability to create/delete events. Notion of Events include the upcoming exams/quiz and latest assignment/project presentation/demo deadlines. Adding or Deleting Students as a user in the course participants. View his/her own Time table for the week/month/year. Ability to create Exams, Submission portals, polls, etc. Analyse *more* in depth course performance with live feedback mechanism. To develop an internal messaging/ discussion forum amongst course participants including the course instructor along with students.

For the *Student* as a User: The ability to view detailed course description of a course. To develop a To do list for monitoring his/her future events in a clear and precise manner. A messaging system, to communicate with co-participants in the course personally and also with the course instructor. Ability to take part in real-time discussion forums.

Ability to observe detailed Time table along with annotations of upcoming events demarcated precisely.

Access to a personal cloud space to store the uploads/Assignments. Access to standardized coding platform. So that all student gets uniform platform.

6 Files and Directories

Posstgresql Tables include, FML, DBMS, DECS, ALGORITHMS, SOFTWARE LAB, USERS, etc. Various php and html files.

Main Files include as follows, admin.php, DBMS 10.php, DECS.php, index.php, SL 10.php, to do.php, ALG 10.php, DBMS.php, FML 10.php, lo-

gout.php, SOFTWARE LAB.php, widget.php, ALGORITHMS.php, DECS 10.php, FML.php, signin.php, table.php.

7 Executing Instructions

Initially database has to be made in postgresql, so within that environment all transactions with database takes place. Just access on any browser in the same local network using localhost/software-lab/darkpan-1.0.0/signin.php or using ip-addr/software-lab/darkpan-1.0.0/signin.php for accessing via a remote machine.